What Survives a Placebo?
Entropy, Market Coupling and the Plausibility of Stress Scenarios

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Acknowledgments. [To be completed.]

Data availability. The data and code that reproduce every table and figure are in the replication package accompanying this article (README file). Sources: Kenneth R. French Data Library (48 industry portfolios), the S&P 500 sample distributed with `skfolio`, the ECB Statistical Data Warehouse and the EBA 2025 stress-test scenario.

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Abstract

Entropy-based stress indicators computed from rolling moments display jumps, slow relaxation and fattening tails in crises. We show that a resampling null with no dynamics reproduces almost all of these signatures: they are estimation artefacts. One fact survives, including a volatility-preserving GARCH null and a within-year comparison, on US stocks, industries and euro-area markets: under stress, correlations couple and the effective number of independent risk sources falls. We give the finite-sample bias of the coupling estimator and the size and power of the test, and price stress scenarios in nats, including the EBA 2025 adverse scenario.

Keywords: entropy; Kullback–Leibler divergence; total correlation; placebo test; systemic risk; stress testing.

JEL classification: C14, C58, G01, G17, G21.

1 Introduction

Information theory offers an appealing vocabulary for financial stress. The Gaussian is the maximum-entropy law for a given mean and covariance, so any departure from it, in shape or in dependence, is information in a precise sense and can be measured in nats. A growing set of indicators builds on this idea, either explicitly through entropy or implicitly through correlation-concentration statistics such as the absorption ratio (Kritzman et al., 2011) or principal-component connectedness (Billio et al., 2012). Plotted through crises, such indicators look compelling: they jump at the onset, decay slowly, and coincide with fat tails and negative skewness.

This paper asks which of these signatures are real. Our answer is uncomfortable: almost none. Every entropy indicator is a convex, non-negative function of estimated moments. On fat-tailed data, an extreme day entering an estimation window raises such an index abruptly, and the index decays slowly as the day loses weight, whether or not any information arrived. We make this precise and test every signature against *placebo nulls*: the identical pipeline, run on resampled data that keep the marginal laws and the same-day cross-section but destroy the dynamics. The approach is the surrogate-data method of Theiler et al. (1992), applied to a full estimation pipeline rather than a single statistic.

Main findings. We decompose the entropy of a return vector exactly into scale, dependence and shape, and study the scale-free part, the *structural index* $\mathcal{J}$, which is the Kullback–Leibler divergence between the return law and independent Gaussians with the same variances (Cardoso, 2003). On twenty S&P 500 stocks (1990–2022) and the 48 French (2026) industry portfolios (1990–2026), the raw data display all the predicted crisis signatures. Against an i.i.d. null that keeps fat tails and same-day correlation, one-sided jumps, fattening tails, the co-movement of tails and correlation and the daily sign of jumps all disappear; several are *stronger* in the shuffled data. What survives is the dependence channel. Under stress, total correlation ("coupling") rises by 0.99 nat on stocks and 3.65 nats on

industries, against 0.22 and 0.58 under the null.

The obvious objection is that the i.i.d. null also destroys volatility clustering, so coupling under stress might be volatility clustering plus the bias of Forbes and Rigobon (2002). We answer it with a second null built by filtered historical simulation (Barone-Adesi et al., 1999): GJR-GARCH volatility (Glosten et al., 1993) with constant conditional correlation (Bollerslev, 1990). This null keeps each asset's volatility dynamics and common volatility bursts, and kills only correlation dynamics. Coupling exceeds it on both panels (none of 999 replications reaches the observed gap), and so do its rank and devolatilised versions; the absorption ratio does not. The same holds on a euro-area panel of 28 series spanning the five segments of the CISS. Because correlations also trend over three decades, we add a within-year comparison with calendar-year fixed effects; the gap shrinks by about 61% but, for the raw and rank measures, still exceeds both nulls on both panels. The second null also changes the verdict elsewhere: the claim that crises are dominated by dependence passes the i.i.d. null but not the GARCH one, so it reflects volatility clustering rather than correlation dynamics.

A Monte Carlo study shows why both nulls are needed. When volatility follows a GARCH process with constant correlation, the i.i.d. null rejects in up to 30% of samples at a nominal 5% level, while the GARCH null keeps its size. When volatility switches between regimes, the GARCH null over-rejects. Each null is reliable where the other is not, and the joint rule (pass both) has the right size for 10 and 20 assets (at most 5.0%) while keeping power against genuine correlation regimes. For 48 assets, the joint rule over-rejects with the plain sample correlation (16% under regime-switching volatility) but not with the Ledoit–Wolf estimator we use (at most 6%, power 84%): shrinkage removes the bias that abrupt volatility regimes induce in large correlation matrices.

Contributions. First, a methodological warning with a remedy: moment-based crisis indicators must be calibrated against a null, and the null must preserve volatility dynamics

when the claim concerns dependence. We give the finite-sample bias of sample coupling under independence (Proposition 3), which makes nats comparable across panels of different size, and the size and power of the test. Second, an empirical fact stated with the right standard of evidence: markets couple under stress beyond volatility, beyond estimation mechanics and beyond secular trends, on three markets. Coupling uses the whole correlation spectrum and passes both nulls where the absorption ratio fails. Third, an application to stress testing. Supervisors must design scenarios that are "severe but plausible" (Basel Committee on Banking Supervision, 2018). Following Breuer et al. (2009); Breuer and Csiszár (2013) and Glasserman and Xu (2014), we measure plausibility by a divergence. We show that the cheapest covariance change achieving a given portfolio severity is rank-one along the portfolio's beta vector (Proposition 4), so uniform volatility dilation, the usual default, costs N times the minimum. Classical scenarios pay 1.5 to 3.7 times the minimal cost for their tail loss. The EBA 2025 adverse scenario costs 10.5 nats, between 2008 and 2022 as they actually happened.

Related literature. Entropy has been used to bound pricing kernels (Stutzer, 1996; Backus et al., 2014) and to measure predictability (Maasoumi and Racine, 2002); we use it as a measurement device for dependence. The decomposition of mutual information into correlation and non-Gaussianity is due to Cardoso (2003); total correlation is due to Watanabe (1960). Asymmetric and crisis correlations are documented by Longin and Solnik (2001) and Ang and Chen (2002), and dynamic-correlation models by Engle (2002); our contribution is a test of that fact that is robust to estimation artefacts and to volatility. Dependence-based systemic-risk indicators include Kritzman et al. (2011), Billio et al. (2012), Diebold and Yılmaz (2014) and the ECB's CISS (Holló et al., 2012), and Giglio et al. (2016) evaluate such measures against macroeconomic outcomes. Finite-sample properties of the log-determinant of sample covariance matrices go back to Bartlett (1950); Cai et al. (2015) study differential-entropy estimation in high dimension. Plausibility constraints for stress scenarios

are developed by Breuer et al. (2009), Breuer and Csiszár (2013), Glasserman and Xu (2014) and Glasserman et al. (2015); the Kullback–Leibler ball is also the model-uncertainty set of Hansen and Sargent (2008).

Section 2 sets out the framework and section 3 the estimators and their artefacts. Section 4 describes the nulls and their size and power. Section 5 presents the data and section 6 the results. Section 7 applies the divergence to stress scenarios. Section 8 concludes. Proofs are in Appendix A; an online appendix contains the dynamic postulates, full tables, the severity ladder and a euro-area application.

2 Framework

Let $r \in \mathbb{R}^N$ be a vector of daily log returns with law p, mean μ and covariance $\Sigma = DRD$, where $D = \operatorname{diag}(\sigma_1, \ldots, \sigma_N)$ and R is the correlation matrix. Let $q = \mathcal{N}(\mu, \Sigma)$ be the Gaussian with the same first two moments, and $q_i = \mathcal{N}(\mu_i, \sigma_i^2)$ its marginals. Entropy $h(p) = -\int p \log p$ and divergences are in nats.

Proposition 1 (Three channels)**.**

$$h(p) = \tfrac{N}{2}\log(2\pi e) + \underbrace{\textstyle\sum_i \log \sigma_i}_{scale} - \underbrace{C(R)}_{dependence} - \underbrace{\mathcal{D}(p)}_{shape}, \tag{1}$$

with coupling $C(R) = -\frac{1}{2}\log\det R \geq 0$ *and shape deficit* $\mathcal{D}(p) = D_{\mathrm{KL}}(p \,\|\, q) \geq 0$*. The structural index*

$$\mathcal{J}(p) = \mathcal{D}(p) + C(R) = D_{\mathrm{KL}}(p \,\|\, \textstyle\prod_i q_i) \geq 0 \tag{2}$$

is zero if and only if returns are independent and Gaussian, and it is invariant to the scale of each asset.

$C(R)$ is the total correlation of the Gaussian with correlation R (Watanabe, 1960); for two assets it reduces to $-\frac{1}{2}\log(1-\rho^2)$. An equivalent reading is the *effective number of risk modes* $N_{\mathrm{eff}} = N\exp(-2C/N)$, the number of independent assets with the same generalised

variance, which equals N under independence and falls as assets couple.[1] The scale channel rises with volatility, so the Gaussian entropy $h(q)$ *increases* in crises, by about 5 nats on our stock panel; the scale-free index $\mathcal{J}$ removes that channel and keeps the two that measure constraint.

Proposition 2 (Exact decomposition by blocks)**.** *For a partition of the assets into blocks $B_1, \ldots, B_K$ with correlation sub-matrices R_{kk},*

$$C(R) = \sum_{k=1}^{K} C(R_{kk}) + I_G(B_1; \ldots; B_K), \qquad I_G = \tfrac{1}{2}\Big(\sum_k \log\det R_{kk} - \log\det R\Big) \geq 0, \quad (3)$$

where I_G is the Gaussian multi-information between blocks, zero if and only if the blocks are mutually uncorrelated.

Proposition 2 turns coupling into a map: how much dependence lies within each market segment and how much between segments, in one unit, for any mix of asset classes. The online appendix uses it to map the coupling of the five segments of the euro-area CISS.

What a stress theory would predict. The framework suggests testable signatures. News lowers expected entropy by exactly the mutual information it carries (Cover and Thomas, 2006, ch. 2), so $\mathcal{J}$ should rise in jumps and relax slowly; jumps should coincide with negative skewness; and because major news hits all assets on the same day, coupling and tail thickness should rise together. Table 1 lists the eight hypotheses we test. The online appendix states the dynamic postulates (an Ornstein–Uhlenbeck process with positive jumps, in the class of Barndorff-Nielsen and Shephard, 2001) and the volume-time argument that motivates the Gaussian as the zero point (Clark, 1973; Tauchen and Pitts, 1983; Ané and Geman, 2000). No result below depends on them.

3 Estimation and its artefacts

3.1 Estimators

Correlations are estimated on a 252-day rolling window with Ledoit and Wolf (2004) shrinkage; for the jump analysis we use an exponentially weighted estimator (60-day half-life, 10% shrinkage to the diagonal), so that an extreme day *leaving* the window does not create a spurious fall. The shape deficit is estimated asset by asset with the approximation of Hyvärinen (1998), whose two terms separate skewness and tails; the Edgeworth expansion $S^2/12+K^2/48$ overstates the deficit of a Student-t_4 by a factor of 69 and is not used. Because $\mathcal{D}$ is estimated marginally, $\mathcal{J}$ omits non-linear dependence and understates the true divergence. A jump in $\mathcal{J}$ is a daily change larger than four times a jump-robust local scale built from lagged bipower variation (Barndorff-Nielsen and Shephard, 2004). The stress regime is the set of dates on which the 21-day volatility of the equal-weight portfolio exceeds the 90% quantile of its past values (expanding window, two-year burn-in), so the threshold never uses future information.

3.2 Finite-sample bias of coupling

Sample coupling is positive even when assets are independent. The classical likelihood-ratio test of sphericity gives its order of magnitude.

Proposition 3 (Bias of sample coupling)**.** *Let $r_1, \dots, r_T$ be i.i.d. $\mathcal{N}(\mu, D^2)$ (independent assets) and $\hat{R}$ the sample correlation matrix, $T > N$. Then $-\Big(T - 1 - \frac{2N+5}{6}\Big) \log\det \hat{R}$ converges in distribution to $\chi^2_{N(N-1)/2}$ as $T \to \infty$ (Bartlett, 1950), and*

$$\mathbb{E}\Big[C(\hat{R})\Big] \;\approx\; b(N,T) = \frac{N(N-1)}{4\Big(T - 1 - (2N+5)/6\Big)}. \tag{4}$$

For our panels the bias of the sample correlation is material: $b(20, 252) = 0.39$ nat and $b(48, 252) = 2.41$ nats, as large as a third of the stress effect on stocks. Table 2 checks

the approximation by simulation. It is accurate for Gaussian data (0.39 for $N = 20$) and equally accurate for independent Student-t_5 data (0.39), because the sampling variance of a correlation between independent variables does not depend on their fourth moments. Under independence, Ledoit and Wolf (2004) shrinkage removes almost all of it (0.002), which is one reason to use the shrunk estimator throughout; when assets are correlated, shrinkage instead biases coupling downward, by an amount that grows with N/T. Two consequences follow. Coupling levels from unshrunk estimators, or from panels of different N, are not comparable without a correction such as $N \exp\{-2[C(\hat{R}) - b]/N\}$ for the effective number of modes. Regime *gaps* compare windows of equal length, so either bias cancels to first order; this is why every test below is on a gap.

3.3 The mechanical sawtooth

The bias is the static part of the artefact; the dynamic part is more damaging. Feed independent Gaussian returns with constant covariance through the pipeline. One injected stress day raises $\mathcal{J}$ by 2.1 nats, and the index decays with a half-life of 35 days purely through the estimator's memory. With no event at all, the jump detector finds 37 positive jumps and no negative one. The reason is convexity: an outlier raises the fourth-moment terms of the shape deficit, and the common component of a large day raises every sample correlation at once; both effects enter immediately and leave at the rate at which the window forgets. The "sawtooth" of rapid rises and slow decays is thus what a fat-tailed law produces once it is passed through a rolling estimator, and it is not evidence of information arriving in bursts.

4 Inference with placebo nulls

4.1 Two nulls

Let $S(X)$ be any statistic computed by the full pipeline on the return panel X (a regime gap, a jump count, a half-life). A null is a resampling scheme X^* that preserves some features

of X and destroys others; the p-value is $\hat{p} = (1 + \#\{b : S(X_b^*) \geq S(X)\})/(B + 1)$ in the predicted direction, with $B = 999$ replications. The stress regime is recomputed on each replication.

i.i.d. null. Rows of X are drawn with replacement. It preserves each day's cross-section (fat tails, same-day correlation, common extreme days) and destroys all dynamics. It answers: beyond the fact that turbulent periods contain big days, does the measure detect any dynamics?

CCC-FHS null. We fit a GJR-GARCH(1,1) by Gaussian quasi-maximum likelihood to each asset, collect the $T \times N$ matrix of standardised residuals $\hat{z}_t$, resample its rows i.i.d., and rebuild returns with the fitted recursions. Each asset keeps its own volatility clustering and leverage effect, all assets share the resampled day (so volatility bursts remain common), and the conditional correlation is constant by construction. It answers the question specific to the dependence channel: is coupling under stress more than what volatility dynamics and the Forbes and Rigobon (2002) effect generate when correlations do not move? Fitted persistences average 0.989 on stocks and 0.987 on industries (Appendix B).

A null we do not use. A circular block bootstrap with 21-day blocks keeps all dynamics shorter than a month. A crisis plays out precisely at that scale (the COVID episode raised $\mathcal{J}$ by 16 nats in three weeks), so a block null keeps much of the phenomenon to be measured, including its correlation dynamics. The CCC-FHS null is the better-targeted alternative: it keeps monthly volatility dynamics but not correlation dynamics.

4.2 Size and power

Table 3 reports a Monte Carlo study of the coupling-gap test with $N = 10$, 20 and 48 assets (the sizes of our panels), $T = 4{,}000$ days and the paper's past-only regime rule, with 1,000, 500 and 300 simulated panels and 199 null replications per test. For speed, the

simulation uses the plain sample correlation on 250-day windows evaluated every fifth day. Under i.i.d. Student-t_5 returns with constant correlation, both nulls have about the right size. Under a constant-correlation GJR-GARCH, the i.i.d. null rejects in 18% of samples with ten assets and 30% with 48: volatility clustering alone creates a spurious coupling gap that a dynamics-free null cannot absorb. The CCC-FHS null keeps its size (5% and 4%). Under Markov-switching volatility with constant correlation, the GARCH null, now misspecified, over-rejects (10% with ten assets). When correlation genuinely rises in the high-volatility regime, from 0.30 to 0.50, both nulls reject often (56% and 73% with twenty assets); a rise to 0.40 is harder to detect (29% and 47%).

Neither null is uniformly valid, but their failures occur under different processes. The paper therefore uses a joint rule: a dependence result is established only if it passes both. With 10 and 20 assets this rule has the right size under all three constant-correlation processes (at most 5.0% at the 5% level) and keeps power against the correlation regime (55% with twenty assets). With 48 assets and the sample correlation, it over-rejects under regime-switching volatility (16%): both nulls miss a bias that abrupt variance shifts create in a large sample correlation matrix, whose finite-sample bias (Proposition 3) grows like N^2/T. The last block of Table 3 reruns the 48-asset case on the same simulated panels with the Ledoit–Wolf estimator of the empirical work: the joint rule then has the right size (at most 6%) and power 84%. For claims about dependence, a dynamics-free null is not enough when volatility clusters, a parametric volatility null is not enough when volatility moves in regimes, and in large cross-sections the estimator must be shrunk; together they are enough in every case we simulate.

5 Data

The main panel is twenty S&P 500 stocks, daily, from January 1990 to December 2022 (8,312 returns), the sample distributed with `skfolio`. Individual firms rather than portfolios avoid

the dependence that aggregation creates mechanically. They are firms that survived the whole period, so the panel has survivorship bias; we therefore treat the 48 French (2026) daily value-weighted industry portfolios (January 1990 to June 2026, about 9,000 days, "Other" excluded), which have none, as the second, independent panel. A result counts only if it holds on both. Section 7 and the online appendix use a euro-area panel of 28 daily series covering the five segments of the CISS (2005–2026) and the EBA 2025 scenario (European Banking Authority, 2025).

6 Results

6.1 Coupling under stress

Table 4 is the paper's central result. Under stress, the coupling of the twenty stocks rises by 0.99 nat, the effective number of modes falling from 13.1 to 11.4. The i.i.d. null produces 0.22 nat and the CCC-FHS null -0.06; none of the 999 replications of either reaches the observed gap. On industries the rise is 3.65 nats against 0.58 and 0.67. On industries the volatility-preserving null thus generates a fifth of the observed gap, more than the i.i.d. null, as the Monte Carlo predicts; on stocks it generates none (its mean gap is slightly negative), plausibly because firm-specific volatility paths, which differ more across twenty firms than across 48 diversified industries, lower measured correlations in turbulent windows.

Three variants address the usual objections. Dividing each return by its RiskMetrics volatility before computing correlations ("devolatilised") removes the Forbes and Rigobon (2002) effect directly; replacing returns by normal scores of their ranks makes the measure insensitive to extreme values. Both exceed both nulls on both panels ($\hat{p} \leq 0.004$). Finally, correlations trend over three decades and stress dates cluster in a few years, so a regime gap can absorb a trend. With calendar-year fixed effects, the raw stock gap falls to 0.38 nat (industries 1.43), but it remains beyond both nulls (CCC-FHS means -0.03 and 0.50, $\hat{p} \leq 0.004$), and so does the rank version ($\hat{p} \leq 0.006$). The devolatilised version is the weakest

within years: it is borderline on stocks ($\hat{p} = 0.052$) and fails on industries ($\hat{p} = 0.117$), so within a year part of the stress effect is a volatility effect. Coupling under stress is therefore mostly a within-year phenomenon, not an artefact of slow trends, and its rank version, which is insensitive to extreme values, is the most robust measure throughout.

Among existing measures, the absorption ratio of Kritzman et al. (2011) fails the i.i.d. null on stocks and the CCC-FHS null on both panels; the first-eigenvalue share passes both, by a smaller margin than coupling. $\log\det R$ uses the whole spectrum, including the small eigenvalues that collapse when assets couple, whereas the absorption ratio uses only the top fifth.

Figure 1 shows the observed gap against the two null distributions, and Figure 2 the coupling series with the stress dates. The euro-area panel of the online appendix gives the same answer: total coupling and coupling between the five CISS segments beat both nulls ($\hat{p} \leq 0.002$).

6.2 What does not survive

Table 5 confronts every hypothesis with both nulls. A result passes if it beats both, after the Holm (1979) step-down adjustment over the 26 tests of the table, applied to the larger of its two p-values (the intersection-union test). With 999 replications the smallest p-value is 0.001, below the first Holm threshold ($0.05/26 = 0.0019$), so the adjustment can be passed by exact rank p-values. 6 results pass: coupling on both panels, the range of $\mathcal{J}$ on stocks, its daily half-life on both panels and the largest episode on stocks. Two more beat both nulls at 5% but not after adjustment: the range of $\mathcal{J}$ and the largest episode on industries. The failures are informative.

One-sided jumps (H3). The detector finds 158 upward and 9 downward jumps on stocks, but the i.i.d. null produces 148 and 3: it is *more* one-sided than the data, and so is the GARCH null (163 and 3). Counting by episode (jumps less than 40 days apart) changes nothing (65 rising against 7 falling, but 73 to 3 under the i.i.d. null). *Shape under stress*

(H1, $\mathcal{D}$). Tails fatten under stress by 0.27 nat, but by 1.24 under the i.i.d. null, because a stress window in a shuffled world contains isolated extreme days; the GARCH null matches the data (0.30). *Co-movement of channels (H6).* Dependence and tails move together under stress (0.63), but the i.i.d. null couples them more (0.90). *Daily sign (H5).* Jumps coincide with negative skewness (correlation -0.24), but the GARCH null, whose leverage effect ties volatility bursts to falling prices, produces more (-0.38). These are exactly the signatures an entropy story of crises predicts, and none is distinguishable from estimation mechanics or from volatility dynamics.

Three dynamic facts pass both nulls. The *range* of $\mathcal{J}$ (19.0 nats on stocks, against 10.0 and 4.8) and its daily *half-life* (121 days against 91 and 50; on industries 190 against 101 and 53) exceed what either null produces, and so does the largest episode on stocks (COVID-19, 15.8 nats against 8.0 and 3.8). Episode by episode, however, relaxation from peak to half-way is no slower than under the i.i.d. null (median 18 days against 24): persistence comes from the succession of turbulent days, not from slower absorption of each shock.

6.3 Crisis composition

We split the rise of $\mathcal{J}$ within each episode exactly into dependence, skewness and tail terms. Skewness almost never dominates. Dependence dominates 46% of the 65 stock episodes and 79% of the 61 industry episodes, far more than under the i.i.d. null (26% and 42%, no replication reaching either). The GARCH null, however, produces as many dependence-dominated episodes (40% and 86%). When volatility clusters and bursts are common to all assets, the large days of an episode share a common component, and the decomposition attributes the episode to dependence even though the correlation is constant. Composition therefore does not separate real crises from volatility clusters; the regime gap of section 6, which compares correlation levels rather than episode budgets, does.

6.4 Forecasting

Does $\mathcal{J}$ improve risk forecasts beyond volatility? We forecast realised volatility, drawdown and worst daily loss at 5, 10, 21 and 63 days with models re-estimated yearly on past data, and evaluate on non-overlapping dates. Table 6 reports the best entropy regressor, an "unhealed-shock" indicator $\mathcal{J}_t / \max_{252} \mathcal{J}$. It reduces the mean squared error of the 5-day volatility forecast by 3.5% on stocks and 5.4% on industries. Because the models are nested, the appropriate test is that of Clark and West (2007). It rejects the null of no predictive content at 5 and 10 days on both panels (one-sided p-values 0.000 and 0.001 on stocks, 0.000 and 0.000 on industries), and at 21 days at the 5% level; the Diebold and Mariano (1995) test, which is undersized for nested models, rejects at 5 days on both panels and at 10 days on industries, and only the industry case at 5 days survives a Bonferroni correction. At 63 days adding entropy *worsens* the forecast on both panels: the population coefficient may not be zero, but it is too small to be worth estimating. The measures are primarily diagnostic, with a genuine but modest predictive content at horizons of one to four weeks.

7 Application: pricing stress scenarios in nats

Supervisors write market scenarios that must be "severe but plausible". In practice severity is measured in losses and plausibility is left to judgement. The divergence of section 2 gives a common unit.

7.1 The unit

For two Gaussians with the same covariance Σ and means differing by Δ, $D_{\mathrm{KL}}(p_1 \,\|\, p_0) = \frac{1}{2}\Delta^\top \Sigma^{-1} \Delta$: a shock of k standard deviations along any direction costs $k^2/2$ nats, and η nats buy $\sqrt{2\eta}$ standard deviations. In general, for Gaussians,

$$D_{\mathrm{KL}}(p_1 \,\|\, p_0) = \tfrac{1}{2}\Big[\Delta^\top \Sigma_0^{-1} \Delta + \operatorname{tr}(\Sigma_0^{-1}\Sigma_1) - N - \log\det(\Sigma_0^{-1}\Sigma_1)\Big], \tag{5}$$

the sum of a location cost and a deformation cost. A scenario of severity η is the most adverse law within η nats of the current one (Breuer and Csiszár, 2013; Glasserman and Xu, 2014). For a portfolio w and a Gaussian reference, it shifts the portfolio mean by u standard deviations and scales its volatility by v, at cost $g(u, v) = \frac{1}{2}(u^2 + v^2 - 1 - 2\log v)$.

7.2 The cheapest way to reach a severity

The portfolio problem fixes the law of $w^\top r$, not how each asset moves. The next result answers that question.

Proposition 4 (Minimal lift)**.** *Let $p_0 = \mathcal{N}(\mu_0, \Sigma_0)$, $s_0^2 = w^\top \Sigma_0 w$ and $\beta = \Sigma_0 w / s_0^2$. Among Gaussian scenarios whose portfolio has mean $w^\top \mu_0 + u s_0$ and standard deviation $v s_0$, the divergence from p_0 is minimised by*

$$\mu_1 = \mu_0 + u\, s_0\, \beta, \qquad \Sigma_1 = \Sigma_0 + (v^2 - 1)\, s_0^2\, \beta\beta^\top, \tag{6}$$

and the minimum equals $g(u, v)$.

Corollary 1 (Uniform dilation)**.** *The uniform dilation $\Sigma_1 = v^2 \Sigma_0$ has deformation cost $N \cdot \frac{1}{2}(v^2 - 1 - 2\log v)$, N times that of the minimal lift.*

The minimal scenario is a one-factor scenario: it scales the volatility of the portfolio factor, leaves idiosyncratic risk unchanged, loads the shock on high-beta assets and raises correlations among them. It is what the empirical section says crises do. Uniform dilation of all volatilities, the usual default, is the opposite: it deforms every eigendirection equally.

7.3 Auditing scenarios

Table 7 prices three classical scenarios on the stock panel: all volatilities multiplied by 1.5, 2 or 3, all correlations forced to 0.70, 0.90 or 0.95, and a directional shock of 2, 3 or 4 standard deviations. Their full cost is 6.8, 10.0 and 20.2 nats. The minimal scenario producing the

same one-day ES_{99} costs 1.85, 5.47 and 13.74 nats (Proposition 4): the classical scenarios pay 3.7, 1.8 and 1.5 times the minimum for their tail loss. Even on the portfolio's own law they pay 15 to 33% too much. The components interact strongly: separately, the directional shock, the dilation and the forced correlations cost 4.5, 16.1 and 8.1 nats in the standard scenario, but 10.0 together, because forcing correlations concentrates risk on the common factor, and dilating that factor is cheap.

7.4 The EBA 2025 adverse scenario

The first year of the 2025 EU-wide adverse scenario lowers EU equities by 50%, raises German and EU long rates by 110 and 168 basis points and leaves the euro's exchange rates unchanged (European Banking Authority, 2025). Read as a mean shift of one-year changes on five equity indices, two ten-year yields and four exchange rates, with Σ the 2004–2024 covariance, it costs 10.5 nats, a 4.6-standard-deviation shock on a single factor (Table 8). It is as improbable as 2008 (11.2) or 2022 (11.9) as they happened, less than the worst observed year (14.4, March 2008 to March 2009) and far more than 2011 (3.1). It relies on dependence: priced variable by variable it would cost 31.9 nats. The covariance is itself a severity choice: with a 2008–2012 crisis covariance the cost falls to 8.4 nats. These figures are an illustration of the audit, not a calibration: the shocks are annual-average deviations and the reference is Gaussian, which overstates costs in the tails.

7.5 Composing a scenario

Because portfolio variance is linear in Σ, any mixture of an "all-volatility" and an "all-correlation" covariance with the same portfolio variance has that variance too. The dependence share α of a scenario can therefore be chosen without changing its severity. Calibrated on the regime gap ($\alpha = 0.14$) or on severe episodes ($\alpha = 0.49$, a descriptive share since episode composition does not separate crises from volatility clusters), the ten-year scenario costs 2.88 and 0.71 nats against 6.80 under uniform dilation, moving towards the minimal

lift (0.41). For a given severity, more dependence means a *less* improbable scenario.

A calibration of the radius η on the market's own 21-day information flow is reported in the online appendix. It is too mild by a factor close to two against realised volatility increases, so we do not attach return periods to radii; the comparisons above are relative and do not depend on it.

8 Conclusion

Entropy indicators of financial stress are easy to build and persuasive to plot. Most of what they show in crises is produced by the estimator: a resampled world without dynamics reproduces the jumps, the slow decays, the fattening tails and the co-movement of tails and correlation, and often exaggerates them. One fact survives a dynamics-free null, a volatility-preserving null and a within-year comparison, on US stocks, industries and euro-area markets: under stress, correlations couple and the number of independent sources of risk falls. Other appealing facts, such as crises being dominated by dependence, pass the first null but not the second. The joint test has correct size when volatility clusters or switches regimes, provided the correlation matrix is shrunk in large cross-sections.

For stress testing, the same divergence prices scenarios in a unit that reads in standard deviations. The cheapest way to reach a severity is a one-factor scenario; uniform dilation costs N times more in deformation, and classical scenarios pay up to 3.7 times the minimum. The EBA 2025 adverse scenario costs as much as 2008 or 2022. Two extensions are natural: a broad survivorship-free panel of individual stocks, and a dynamic-correlation null fitted to calm periods only, which would ask whether crisis coupling exceeds what ordinary correlation dynamics produce.

A Proofs

Proposition 1. For $q = \mathcal{N}(\mu, \Sigma)$, $h(q) = \frac{1}{2}\log((2\pi e)^N \det \Sigma)$ and $\det \Sigma = (\prod_i \sigma_i)^2 \det R$. Because $\log q$ is quadratic, $\mathbb{E}_p[\log q] = \mathbb{E}_q[\log q]$, so $D_{\mathrm{KL}}(p \,\|\, q) = h(q) - h(p)$, which gives (1). Next, $D_{\mathrm{KL}}(p \,\|\, \prod_i q_i) = -h(p) + \sum_i \log \sigma_i + \frac{N}{2}\log(2\pi e)$, and substituting (1) gives $\mathcal{D} + C$. $C \geq 0$ because the eigenvalues of R sum to N, so their geometric mean is at most one. Equality in $\mathcal{J} \geq 0$ holds iff $p = \prod_i q_i$. Scale invariance holds because $\mathcal{D}$ and R are invariant to $r_i \mapsto a_i r_i$, $a_i > 0$.

Proposition 2. By definition, $C(R) - \sum_k C(R_{kk}) = \frac{1}{2}(\sum_k \log\det R_{kk} - \log\det R)$, which is the divergence between $\mathcal{N}(0, R)$ and $\mathcal{N}(0, \mathrm{diag}(R_{11}, \ldots, R_{KK}))$, hence non-negative (Fischer's inequality), and zero iff R is block diagonal.

Proposition 3. Under independence, $\hat{R}$ has the distribution of the sample correlation of N independent Gaussian samples, and the likelihood-ratio statistic for $H_0 : R = I$ is $-T \log\det \hat{R}$. Bartlett's correction $T \mapsto T - 1 - (2N+5)/6$ makes it asymptotically $\chi^2_{N(N-1)/2}$ with error $O(T^{-2})$ in the mean. Taking expectations, $\mathbb{E}[-\log\det \hat{R}] \approx N(N-1)/(2(T-1-(2N+5)/6))$; halve it. For non-Gaussian independent data the leading term is unchanged as long as fourth moments are finite, because $\sqrt{T}\hat{\rho}_{ij} \to \mathcal{N}(0,1)$ for independent pairs; Table 2 shows the approximation for t_5.

Proposition 4. By the data-processing inequality (Cover and Thomas, 2006, ch. 2), every scenario costs at least the divergence of its portfolio law, $g(u, v)$. It remains to show that (6) attains it. $w^\top \beta = 1$, so the portfolio mean and variance are as required. In (5), $\Delta^\top \Sigma_0^{-1} \Delta = u^2 s_0^2 \beta^\top \Sigma_0^{-1} \beta = u^2$; $\mathrm{tr}(\Sigma_0^{-1}\Sigma_1) = N + v^2 - 1$; and by the matrix determinant lemma $\det(\Sigma_0^{-1}\Sigma_1) = v^2$. Hence the divergence is $g(u, v)$.

Corollary 1. All eigenvalues of $\Sigma_0^{-1}\Sigma_1$ equal v^2; apply (5).

B Estimation details

GJR-GARCH fits. Each asset's percentage log return follows $r_t = \mu + \varepsilon_t$, $\varepsilon_t = \sqrt{h_t} z_t$, $h_t = \omega + (\alpha + \gamma \mathbf{1}\{\varepsilon_{t-1} < 0\})\varepsilon_{t-1}^2 + \beta h_{t-1}$, estimated by Gaussian QMLE. Median parameters: stocks $\alpha = 0.033$, $\gamma = 0.063$, $\beta = 0.920$; industries $\alpha = 0.030$, $\gamma = 0.079$, $\beta = 0.916$. Simulated paths start at the unconditional variance, with persistence capped at 0.999. Standardised residuals are re-centred and re-scaled before resampling.

Regime gaps and inference. The stress-minus-calm gap compares the mean of a rolling statistic on stress dates and on calm dates (all other dates after the burn-in). The within-year gap is the coefficient of a stress dummy in a regression of the statistic on the dummy and calendar-year dummies. Two consecutive windows share 251 of 252 days, so naive standard errors are meaningless; all inference uses the nulls. Regime comparisons in the online appendix use a circular block bootstrap with 252-day blocks.

Forecasting. Expanding-window OLS, refitted every 252 days, evaluated on dates spaced by the horizon; HAC lag $\max(h, 5)$. The Clark–West statistic uses the adjusted differential $(y - f_1)^2 - [(y - f_2)^2 - (f_1 - f_2)^2]$, with f_1 the volatility-only forecast.

Notes

[1] N_{eff} is N times the geometric mean of the eigenvalues of R. It is a monotone transform of C for fixed N, so it inherits every test of C.

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Table 1: Hypotheses tested

	Statement
H1	*Regime signature.* Under stress, coupling rises, the shape deficit rises, hence $\mathcal{J}$ rises.
H2	*Compensation.* When volatilities rise, dependence absorbs part of the entropy increase.
H3	*One-sided jumps.* Large moves of $\mathcal{J}$ are rises; rising episodes are more numerous and larger.
H4	*Relaxation.* After an episode, $\mathcal{J}$ decays with a half-life longer than the estimator's memory.
H5	*Sign.* Jumps and rising episodes coincide with negative market skewness.
H6	*Channel co-movement.* Dependence and tails move independently in calm markets and together under stress.
H7	*Forecasting.* Entropy improves out-of-sample short-term risk forecasts beyond volatility.
H8	*Crisis composition.* In crisis episodes, dependence is the dominant channel more often than without dynamics.

H3, H4, H5 and H8 concern jumps of $\mathcal{J}$; they are tested day by day and by episode (jumps less than 40 days apart).

Table 2: Finite-sample bias of sample coupling under independence

N	Window T	Law	$b(N,T)$, eq. (4)	Sample correlation	Ledoit–Wolf
10	252	Gaussian	0.091	0.091 (0.019)	0.001 (0.003)
10	252	Student-t_5	0.091	0.091 (0.019)	0.001 (0.003)
20	252	Gaussian	0.390	0.389 (0.039)	0.002 (0.004)
20	252	Student-t_5	0.390	0.390 (0.040)	0.002 (0.004)
48	252	Gaussian	2.409	2.418 (0.102)	0.002 (0.004)
48	252	Student-t_5	2.409	2.404 (0.105)	0.002 (0.004)
20	504	Gaussian	0.192	0.191 (0.019)	0.001 (0.002)
20	504	Student-t_5	0.192	0.192 (0.020)	0.001 (0.002)

Mean (standard deviation) of $C(\hat{R}) = -\frac{1}{2}\log\det\hat{R}$ in nats over 2,000 simulated windows (500 for $N = 48$) of independent assets. Student-t_5 draws are independent across assets. The Ledoit–Wolf column applies the shrinkage used in the empirical work.

Table 3: Size and power of the coupling test: Monte Carlo

Data-generating process		Mean gap (nats)	Rejection at 5% i.i.d.	Rejection at 5% CCC-FHS	Rejection at 5% both	Rejection at 1% i.i.d.	Rejection at 1% CCC-FHS	Rejection at 1% both
$N = 10$ *assets, 1,000 panels per process*								
i.i.d. Student-t_5, constant correlation 0.30	size	0.11	0.048	0.047	0.040	0.005	0.005	0.002
GJR-GARCH(1,1), t_5 shocks, constant correlation 0.30	size	0.12	0.176	0.048	0.047	0.061	0.004	0.004
Markov-switching volatility (×2.5), constant correlation	size	0.12	0.048	0.102	0.046	0.014	0.027	0.011
Markov-switching volatility, correlation 0.30 → 0.40 in stress	power	0.24	0.200	0.361	0.196	0.068	0.152	0.056
Markov-switching volatility, correlation 0.30 → 0.50 in stress	power	0.40	0.500	0.718	0.495	0.260	0.470	0.242
$N = 20$ *assets, 500 panels per process*								
i.i.d. Student-t_5, constant correlation 0.30	size	0.24	0.046	0.048	0.036	0.006	0.002	0.002
GJR-GARCH(1,1), t_5 shocks, constant correlation 0.30	size	0.33	0.218	0.038	0.036	0.058	0.004	0.004
Markov-switching volatility (×2.5), constant correlation	size	0.31	0.068	0.110	0.050	0.010	0.020	0.004
Markov-switching volatility, correlation 0.30 → 0.40 in stress	power	0.62	0.292	0.468	0.276	0.124	0.242	0.104
Markov-switching volatility, correlation 0.30 → 0.50 in stress	power	0.94	0.560	0.732	0.552	0.324	0.462	0.294
$N = 48$ *assets, 300 panels per process*								
i.i.d. Student-t_5, constant correlation 0.30	size	0.65	0.033	0.023	0.023	0.013	0.013	0.013
GJR-GARCH(1,1), t_5 shocks, constant correlation 0.30	size	1.03	0.303	0.037	0.033	0.117	0.007	0.003
Markov-switching volatility (×2.5), constant correlation	size	1.23	0.183	0.243	0.157	0.060	0.110	0.047
Markov-switching volatility, correlation 0.30 → 0.40 in stress	power	1.97	0.447	0.537	0.423	0.207	0.273	0.170
Markov-switching volatility, correlation 0.30 → 0.50 in stress	power	2.66	0.653	0.740	0.620	0.380	0.457	0.337
$N = 48$ *assets, Ledoit–Wolf estimator (the paper's), 200 panels, 99 null replications*								
i.i.d. Student-t_5, constant correlation 0.30	size	0.39	0.045	0.045	0.040	0.010	0.005	0.005
GJR-GARCH(1,1), t_5 shocks, constant correlation 0.30	size	0.22	0.075	0.060	0.050	0.025	0.020	0.010
Markov-switching volatility (×2.5), constant correlation	size	0.26	0.060	0.115	0.060	0.020	0.030	0.015
Markov-switching volatility, correlation 0.30 → 0.50 in stress	power	1.93	0.835	0.925	0.835	0.565	0.780	0.555

$T = 4{,}000$ days; coupling of the sample correlation on 250-day windows evaluated every fifth day; stress = past-only 90% quantile of the 21-day volatility of the equal-weight portfolio. Each test uses 199 null replications; the CCC-FHS null re-estimates the GJR-GARCH models on each simulated panel. "Both": rejection by both nulls, the rule used in the paper. Mean gap: stress-minus-calm coupling, positive when assets couple under stress. Upper blocks: plain sample correlation, which the Monte Carlo uses for speed; last block: the Ledoit–Wolf estimator of the empirical work, on the same simulated panels. Monte Carlo standard error of a rejection rate near 5%: 0.7 points with 1,000 panels, 1.0 with 500, 1.3 with 300, 1.5 with 200.

Table 4: Coupling under stress against two placebo nulls

	S&P 500, 20 stocks					48 industries				
		i.i.d. null		CCC-FHS null			i.i.d. null		CCC-FHS null	
Measure	Obs.	mean	p	mean	p	Obs.	mean	p	mean	p
Coupling, raw	0.99	0.22	<0.001	-0.06	<0.001	3.65	0.58	<0.001	0.67	<0.001
Coupling, devolatilised	0.69	0.26	<0.001	0.28	0.003	2.58	0.75	<0.001	1.34	0.004
Coupling, rank	0.80	0.17	<0.001	0.24	<0.001	3.15	0.56	<0.001	1.13	<0.001
Absorption ratio	0.012	0.008	0.270	0.037	0.751	0.025	0.006	<0.001	0.021	0.305
First-eigenvalue share	0.063	0.019	<0.001	-0.000	<0.001	0.048	0.019	<0.001	0.014	<0.001
Within-year gap (calendar-year fixed effects)										
Coupling, raw	0.38	0.11	<0.001	-0.03	<0.001	1.43	0.31	<0.001	0.50	0.004
Coupling, devolatilised	0.24	0.14	0.017	0.14	0.052	0.90	0.43	0.003	0.65	0.117
Coupling, rank	0.36	0.09	<0.001	0.13	0.002	1.28	0.30	<0.001	0.60	0.006

Stress-minus-calm gap of each measure (coupling in nats, signed so that a positive value means more coupling under stress; absorption ratio and first-eigenvalue share in variance shares). Stress: past-only 90% quantile of 21-day equal-weight volatility. p: $(1 + k)/(1 + B)$ with k the number of the $B = 999$ replications at least as extreme in the predicted direction; < 0.001 means no replication reached the observed value. Devolatilised: returns divided by their RiskMetrics volatility. Rank: normal scores. CCC-FHS: constant-correlation GJR-GARCH filtered historical simulation.

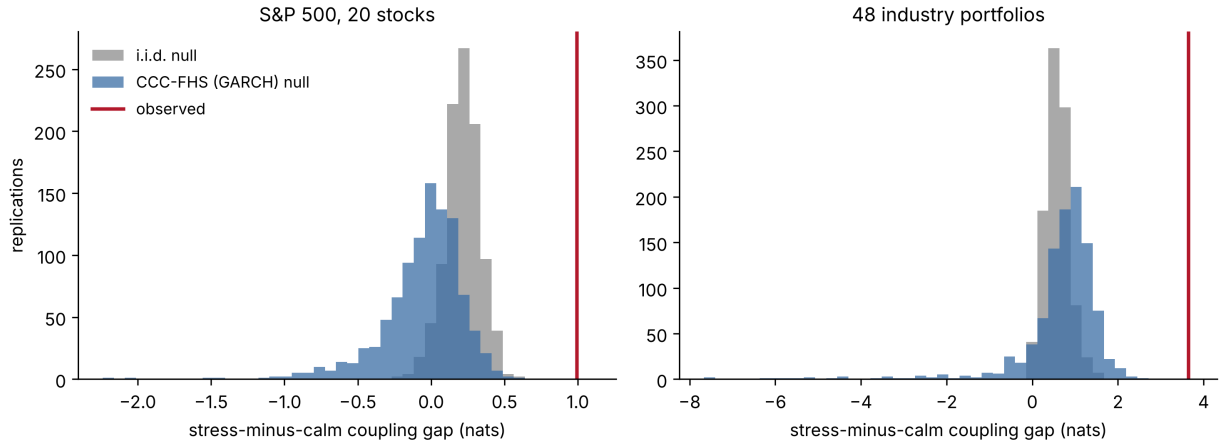


Figure 1: Coupling under stress against two placebo nulls. Each panel shows the stress-minus-calm gap in coupling C (nats; positive means assets couple under stress) for the observed data (vertical line) and for 999 replications of the i.i.d. null and of the CCC-FHS (constant-correlation GJR-GARCH) null. Left: twenty S&P 500 stocks, 1990–2022. Right: 48 industry portfolios, 1990–2026.

Alt text: Two histograms. In each, the two null distributions sit well to the left of a vertical line marking the observed coupling gap, the GARCH null slightly to the right of the i.i.d. null.

Table 5: Verdicts against both placebo nulls with multiple-testing control

				i.i.d. null		CCC-FHS null			
	Statistic	Panel	Observed	mean	p	mean	p	Holm p	Verdict
H1	coupling gap (nats)	stocks	0.99	0.22	<0.001	-0.06	<0.001	0.026	**passes**
H1	coupling gap (nats)	industries	3.65	0.58	<0.001	0.67	<0.001	0.026	**passes**
H1	index $\mathcal{J}$ gap (nats)	stocks	1.93	2.13	0.698	0.81	0.003	1.000	fails
H1	index $\mathcal{J}$ gap (nats)	industries	5.93	8.04	0.971	3.54	0.016	1.000	fails
H1	shape $\mathcal{D}$ gap (nats)	stocks	0.27	1.24	1.000	0.30	0.554	1.000	fails
H1	shape $\mathcal{D}$ gap (nats)	industries	0.98	4.86	1.000	1.04	0.461	1.000	fails
H2	compensation slope β (stress)	stocks	0.20	0.06	0.415	-0.20	0.012	1.000	fails
H2	compensation slope β (stress)	industries	0.23	0.09	0.055	0.12	0.396	1.000	fails
H3	upward jumps (count)	stocks	158	148	0.136	163	0.728	1.000	fails
H3	upward jumps (count)	industries	182	182	0.527	193	0.888	1.000	fails
H3	largest episode (nats)	stocks	15.8	8.0	<0.001	3.8	<0.001	0.026	**passes**
H3	largest episode (nats)	industries	46.3	30.9	0.010	15.8	0.003	0.190	not after Holm
H4	half-life of $\mathcal{J}$ (days)	stocks	121	91	0.002	50	<0.001	0.042	**passes**
H4	half-life of $\mathcal{J}$ (days)	industries	190	101	<0.001	53	<0.001	0.026	**passes**
H4	range of $\mathcal{J}$ (nats)	stocks	19.0	10.0	<0.001	4.8	<0.001	0.026	**passes**
H4	range of $\mathcal{J}$ (nats)	industries	54.5	37.3	0.004	18.5	<0.001	0.080	not after Holm
H4	median half-life from peak (days)	stocks	18	24	0.963	15	0.056	1.000	fails
H4	median half-life from peak (days)	industries	25	25	0.551	17	<0.001	1.000	fails
H5	corr($\Delta\mathcal{J}, \Delta S$), daily	stocks	-0.24	-0.03	0.107	-0.38	0.948	1.000	fails
H5	corr($\Delta\mathcal{J}, \Delta S$), daily	industries	-0.44	-0.21	0.076	-0.52	0.824	1.000	fails
H5	episodes with falling skewness	stocks	0.62	0.50	0.006	0.60	0.375	1.000	i.i.d. only
H5	episodes with falling skewness	industries	0.57	0.53	0.160	0.68	0.985	1.000	fails
H6	channel co-movement gap	stocks	0.59	0.76	0.992	0.42	0.140	1.000	fails
H6	channel co-movement gap	industries	0.41	0.64	1.000	0.53	0.843	1.000	fails
H8	episodes dominated by dependence	stocks	0.46	0.26	<0.001	0.40	0.168	1.000	i.i.d. only
H8	episodes dominated by dependence	industries	0.79	0.42	<0.001	0.86	0.956	1.000	i.i.d. only

999 replications of each null. p: exact rank p-value $(1 + k)/(1 + B)$, one-sided in the direction the hypothesis predicts. Holm p: the Holm (1979) step-down adjustment over the 26 tests of the table, applied to $\max(p_{\text{i.i.d.}}, p_{\text{CCC-FHS}})$, the p-value of the test that rejects only when both nulls do. "Passes": Holm $p \leq 0.05$. "i.i.d. only": beats the i.i.d. null but not the volatility-preserving one. "Not after Holm": beats both nulls at 5% but not after the adjustment. "Null stronger": both nulls produce a larger effect than the data ($p > 0.95$). The coupling gap is in nats, positive for coupling.

Table 6: Out-of-sample forecasts of realised volatility

Horizon (trading days)	5	10	21	63
S&P 500, 20 stocks	-3.5%	-3.5%	-3.3%	+1.0%
DM p (two-sided)	0.029	0.085	0.226	0.999
CW p (one-sided)	0.000	0.001	0.020	0.047
48 industries	-5.4%	-5.2%	-4.2%	+4.2%
DM p (two-sided)	0.001	0.009	0.090	0.161
CW p (one-sided)	0.000	0.000	0.012	0.585

Change in the mean squared error of the realised-volatility forecast when the unhealed-shock indicator $\mathcal{J}_t / \max_{252} \mathcal{J}$ is added to log volatility (negative = better). Expanding-window OLS refitted yearly; evaluation dates spaced by the horizon. DM: Diebold and Mariano (1995); CW: Clark and West (2007), the test for nested models.

Table 7: Cost in nats of classical stress scenarios

Classical scenario	moderate	standard	severe
volatility multiplier	1.5	2.0	3.0
all correlations forced to	0.70	0.90	0.95
directional shock	-2σ	-3σ	-4σ
one-day ES_{99}	9.1%	13.7%	20.3%
full cost (20 assets)	6.80	9.97	20.19
cost seen by the portfolio	2.75	7.24	16.22
minimal cost for the same ES_{99} (Proposition 4)	**1.85**	**5.47**	**13.74**
ratio full / minimal	3.7	1.8	1.5
Cost of each component alone			
directional shock	2.00	4.50	8.00
volatility dilation	4.39	16.14	58.03
forced correlations	2.43	8.10	13.50

Equal-weight portfolio of the twenty stocks; current state: annual volatility 20%, one-day ES_{99} 3.4%. The full cost is the Gaussian divergence (5) on the 20 assets; the cost seen by the portfolio is the divergence of the portfolio's law. The minimal cost is the portfolio divergence $g(u, v)$ of the entropy-ball scenario with the same ES_{99}, attained in full dimension by the rank-one lift (6). The largest 21-day information flow observed in 33 years is about 2 nats.

Table 8: Cost in nats of the EBA 2025 adverse scenario and of observed years

	Equities and rates	With FX fixed
EBA 2025 adverse scenario, year 1	**10.5**	**11.1**
equities only / rates only	6.8 / 3.1	
sum of variable-by-variable costs	31.9	
with a 2008–2012 crisis covariance	8.4	9.0
Observed year 2011 / 2008 / 2022	3.1 / 11.2 / 11.9	3.1 / 18.2 / 13.2
Worst rolling year (March 2008–March 2009)	14.4	19.5

Five equity indices (DAX, CAC 40, FTSE MIB, IBEX 35, AEX), two ten-year euro-area yields (AAA and all ratings) and four euro exchange rates; annual covariance estimated on daily changes from September 2004 to December 2024. Cost $\frac{1}{2}\Delta^\top\Sigma^{-1}\Delta$ of the year-1 deviations from baseline (equities -50%, rates +110 and +168 bp, FX unchanged). Computed with `scripts/eba_nats.py` on the euro-area panel; carried over from the central-bank version of this work.

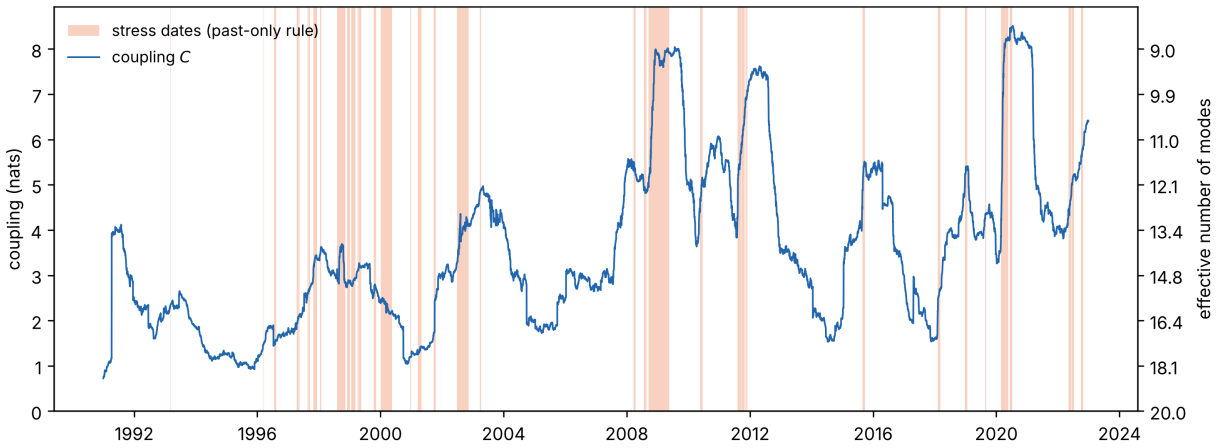


Figure 2: Coupling of twenty S&P 500 stocks, 1990–2022. Rolling 252-day coupling $C = -\frac{1}{2}\log\det R$ (nats, Ledoit–Wolf estimator), with past-only stress dates shaded. The right axis gives the effective number of risk modes $N\exp(-2C/N)$.
Alt text: A time series of coupling with peaks in 2008 and 2020, aligned with shaded stress periods.